Worksheet 3 · Linear Equations IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Solve linear equations with unknowns on one or both sides • Solve equations involving brackets and fractions • Form and solve equations from a given context		
Grade: / 51	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Linear Equations*

Question 1 (7 marks)

Solve $y = \frac{2y+1}{5}$

Show clear algebraic working.

$y =$ .....

(3)

Question 2 (2 marks)

Solve $4x - 13 = 17 + 8x$

$x =$ .....

Question 3 (4 marks)

Solve $\frac{4m+9}{3} = 7 - 2m$

Show clear algebraic working.

$m =$ .....

(4)

Question 4 (3 marks)

Solve $5(x+7) = 2x - 10$

Show clear algebraic working.

$x =$ .....

(3)

Question 5 (4 marks)

Solve $3(2x-5) = \frac{9-x}{2}$

Show clear algebraic working.

$x =$ .....

(4)

Question 6 (4 marks)

Three tins, A, B and C, each contain buttons.

Tin A contains x buttons.
Tin B contains 4 times the number of buttons that tin A contains.
Tin C contains 7 fewer buttons than tin A.

The total number of buttons in the three tins is 137

Work out the number of buttons in tin C.

Question 7 (3 marks)

(a) Solve $p = \frac{3p-5}{10}$

Show clear algebraic working.

$p =$ .....

(3)

Question 8 (3 marks)

Solve $\frac{6x-5}{2} = x + 1$

Show clear algebraic working.

$x =$ .....

(3)

Question 9 (3 marks)

Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x =$ .....

(3)

Question 10 (3 marks)

Solve $5(x + 3) = 3x - 4$

Show clear algebraic working.

Question 11 (4 marks)

The diagram shows the triangle PQR.

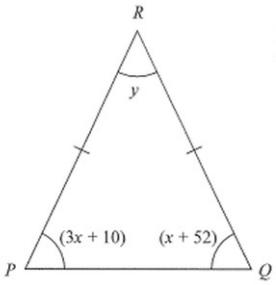


Diagram **NOT** accurately drawn

In the diagram, all the angles are in degrees.

$RP = RQ$

Find the value of y.
Show clear algebraic working.

Question 12 (3 marks)

Solve $4 - 3x = \dfrac{5 - 8x}{4}$

Show clear algebraic working.

Question 13 (4 marks)

The diagram shows a regular hexagon, $ABCDEF$, and an isosceles triangle, GHJ.

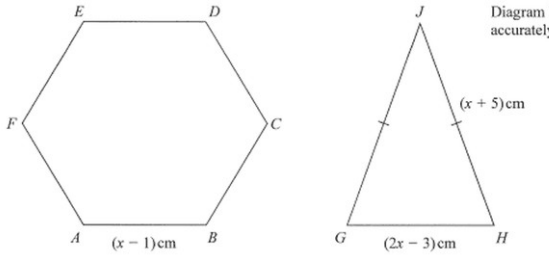


Diagram **NOT** accurately drawn

The perimeter of the hexagon is equal to the perimeter of the triangle.

Find the length of each side of the hexagon.
Show clear algebraic working.

Question 14 (4 marks)

Here are two rectangles, rectangle A and rectangle B.

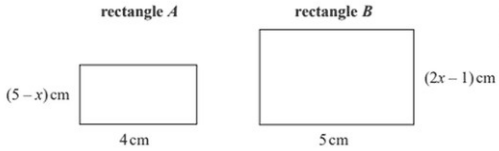


Diagram **NOT** accurately drawn

The area of rectangle B is twice the area of rectangle A.

Work out the value of x.
Show your working clearly.